

GRADE 10 BIOLOGY

LEARNER'S RESOURCE PAMPHLET

*Your quick-reference guide to Cell Biology, Plant Anatomy,
and Animal Anatomy & Physiology*

Based on the KICD Senior School Curriculum Design
For learners aged 14–16

How To Use This Pamphlet

This pamphlet is organised into the three strands you'll meet in Grade 10 Biology. Each topic gives you:

- A short, simple explanation of the concept.
- Clearly organised key facts and lists.
- 💡 DID YOU KNOW? boxes with fun science facts.
- ✎ TRY THIS simple, safe activities and experiments.
- ★ QUICK TIP boxes with memory aids and exam tips.
- 🔑 KEY WORDS you need to know.

Use it alongside your class notes, textbook, and laboratory work — this pamphlet is a quick companion, not a replacement for full lessons and practicals.

Contents

- STRAND 1 — Cell Biology and Biodiversity (4 topics)
- STRAND 2 — Anatomy and Physiology of Plants (3 topics)
- STRAND 3 — Anatomy and Physiology of Animals (3 topics)

STRAND 1

CELL BIOLOGY AND BIODIVERSITY

This strand introduces you to Biology as a subject, teaches you how scientists collect and study living things, and takes you deep inside the cell — the basic unit of all life.

1.1 Introduction to Biology

Biology is the study of life — everything from tiny bacteria to blue whales. Understanding Biology helps you make informed choices about your health, food, and environment, and opens the door to many exciting careers.

Fields of study in Biology

- Botany — the study of plants
- Zoology — the study of animals
- Taxonomy — classifying living organisms
- Anatomy — study of body structure
- Physiology — study of how body parts function
- Ecology — study of organisms and their environment
- Biochemistry — the chemistry of living things
- Biotechnology — using living organisms/systems to develop products
- Genetics — the study of heredity
- Microbiology — study of microorganisms
- Parasitology & Entomology — study of parasites and insects

Careers linked to Biology

Medicine, nursing, pharmacy, agriculture, veterinary science, environmental conservation, food technology, forensic science, research, and teaching — Biology is a foundation for many rewarding careers.



DID YOU KNOW?

Career choices should be based on your interests and abilities — not on gender, culture, disability, or stereotypes. Anyone can pursue any Biology-related career!

1.2 Specimen Collection and Preservation

Biologists collect, process, and preserve specimens so they can be studied closely, now or in the future.

Common apparatus used

- Pooter/aspirator — for collecting small insects
- Pitfall trap — a buried container that traps ground-crawling insects
- Sweep net/aerial net — for catching flying insects
- Light trap — attracts night-flying insects using light
- Tullgren funnel — extracts small organisms from soil/leaf litter using heat and light
- Hand lens, forceps, collecting bags, labels, and preservation jars

Processing and preserving specimens

- A herbarium preserves plant specimens by pressing, drying, mounting, and labelling them.
- Animal specimens may be preserved wet (in ethanol) or dry (mounted on soft boards).
- Every specimen should be labelled with its common/local name, date, and place of collection.

★ QUICK TIP

Always collect only the specimens you need, handle organisms humanely, and observe safety precautions (gloves, careful handling of traps) — conservation starts with responsible collection.

1.3 Cell Structure and Specialization

The cell is the basic unit of life. Every living thing is made of one or more cells, and understanding their structure helps explain how living things work.

Light microscope vs electron microscope

- A light microscope uses visible light and lenses; it's affordable and common in schools, but has lower magnification and resolution.
- An electron microscope uses a beam of electrons, giving much higher magnification and resolution — it can reveal tiny organelles invisible under a light microscope.
- Magnification = how much bigger an image appears; resolution = how clearly fine detail can be seen.

Plant cells vs animal cells

- Plant cells have a cell wall (made of cellulose), a large permanent vacuole, and chloroplasts (for photosynthesis).
- Animal cells lack a cell wall and chloroplasts, and usually have small, temporary vacuoles.
- Both have a cell membrane, cytoplasm, nucleus, mitochondria, and ribosomes.

Specialized cells and their adaptations

- Root hair cell (plant): long, thin extension increases surface area for water/mineral absorption.
- Palisade cell (plant): packed with chloroplasts near the leaf surface for maximum light absorption.
- Guard cell (plant): controls opening/closing of stomata for gas exchange.
- Nerve cell (animal): long fibre (axon) for fast transmission of electrical impulses.
- Red blood cell (animal): biconcave shape and no nucleus, maximising space to carry oxygen.
- Muscle cell (animal): contains many mitochondria and contractile fibres for movement.
- Sperm cell (animal): has a tail (flagellum) for swimming towards the egg.

Levels of organisation

Cells → Tissues (groups of similar cells) → Organs (groups of tissues) → Organ systems (groups of organs working together) → Organism.

TRY THIS

Prepare a wet-mount slide of onion epidermis. Observe it under a light microscope and sketch what you see, labelling the cell wall, cytoplasm and nucleus.

1.4 Chemicals of Life

Living things are built from — and run on — a set of essential chemicals. Knowing what they do (and how to test for them) is central to Biology.

Major chemicals of life

- Carbohydrates: main energy source (e.g. starch, sugars); found in foods like rice, ugali, bread.
- Lipids (fats/oils): energy storage, insulation, and cell membrane structure.
- Proteins: growth and repair of body tissues, and forming enzymes.
- Vitamins: small amounts needed for specific body functions (e.g. Vitamin C for healthy skin and immunity).
- Water: universal solvent, transport medium, and temperature regulator in the body.
- Mineral salts: needed for processes like bone formation (calcium) and oxygen transport (iron).

Food tests you should know

- Iodine test — for starch (turns blue-black).
- Benedict's test — for reducing sugars (turns brick-red on heating if present).
- Biuret test — for proteins (turns purple/violet).
- Emulsion/grease-spot test — for lipids (translucent spot on paper).
- DCPIP test — for vitamin C (blue dye decolourises).

Enzymes

Enzymes are biological catalysts — they speed up chemical reactions in cells without being used up. Catalase, for example, breaks down hydrogen peroxide in living tissues.

- Factors affecting enzyme activity: temperature, pH, substrate concentration, and enzyme concentration.
- Very high temperatures denature enzymes, stopping them from working.



KEY WORDS

Catalyst: a substance that speeds up a reaction without being consumed.

Denature: to permanently change an enzyme's shape so it stops working.

Substrate: the substance an enzyme acts upon.



DID YOU KNOW?

Next time you buy packaged food, check the label — it lists the chemicals of life it contains, plus preservatives, colourings, and an expiry date. Reading labels helps you make healthier choices!

STRAND 2

ANATOMY AND PHYSIOLOGY OF PLANTS

Plants may look still and simple, but inside they run complex systems for feeding, transporting materials, and exchanging gases. This strand explores how plants stay alive and thrive.

2.1 Nutrition in Plants

Plants get their food in different ways. Most make their own food, but some depend on other organisms.

Types of nutrition

- Autotrophic nutrition: making your own food, e.g. through photosynthesis (most green plants).
- Heterotrophic nutrition: depending on other organisms for food, including:
 - Parasitic — feeding off a living host, e.g. dodder plant
 - Saprophytic — feeding on dead/decaying matter, e.g. some fungi
 - Symbiotic — a partnership where both organisms benefit, e.g. mycorrhizae with plant roots
 - Insectivorous — trapping and digesting insects for extra nutrients, e.g. Venus flytrap

The Chloroplast

The chloroplast is the organelle where photosynthesis happens. It contains chlorophyll (the green pigment that absorbs light energy) arranged on membranes called thylakoids, with a fluid-filled stroma where sugar-making reactions occur.

Photosynthesis: the light and dark stages

- Light stage (light-dependent reactions): occurs on the thylakoid membranes; light energy splits water and produces energy carriers (ATP) and oxygen (released as a by-product).
- Dark stage (light-independent reactions/Calvin cycle): occurs in the stroma; uses the energy carriers from the light stage to combine carbon dioxide into glucose.

Overall equation:

Carbon dioxide + Water → (light, chlorophyll) → Glucose + Oxygen



DID YOU KNOW?

Nearly all oxygen in the atmosphere is a by-product of photosynthesis — plants (and algae) have literally shaped the air we breathe over billions of years.

2.2 Transport in Plants

Plants need to move water, minerals, and manufactured food between their roots, stems, and leaves — this is done by specialised vascular tissues.

Vascular tissues

- Xylem: transports water and mineral salts upward from roots to leaves; also provides structural support.
- Phloem: transports manufactured food (mainly sugars) from the leaves to other parts of the plant (translocation).
- In monocotyledonous plants, vascular bundles are scattered throughout the stem; in dicotyledonous plants, they are arranged in a distinct ring.

Uptake of water and minerals

- Root pressure: pushes water up from the roots.
- Capillarity: water rises in narrow xylem tubes due to surface tension.
- Transpiration pull: as water evaporates from the leaves, it pulls more water up the plant — the main driving force of water transport.

Transpiration

Transpiration is the loss of water vapour from plant surfaces, mainly through leaf stomata.

- Factors that increase transpiration: high temperature, low humidity, strong wind, and bright light.
- Factors that decrease transpiration: high humidity, still air, low light, and low temperature.



TRY THIS

Cover a leafy plant's leaf with a clear plastic bag and leave it in sunlight for an hour. Observe the water droplets that collect inside — that's transpired water vapour condensing!

Translocation

Translocation is the movement of manufactured food (sugars) through the phloem from where it's made (the leaves) to where it's needed or stored (roots, fruits, growing parts).



QUICK TIP

A bark-ringing (girdling) experiment — carefully removing a ring of bark (which contains phloem) from a tree branch — causes food to build up above the ring, proving that phloem transports food downward. Never do this without guidance, as it can seriously harm the tree.

2.3 Gaseous Exchange and Respiration in Plants

Plants exchange gases with their environment (taking in and giving out oxygen and carbon dioxide) and use respiration to release energy from food.

Sites of gaseous exchange

- Stomata: tiny pores, mainly on leaf surfaces, that open and close to control gas exchange and water loss.
- Lenticels: small openings on woody stems for gas exchange.
- Pneumatophores: special aerial roots in some aquatic/swamp plants (e.g. mangroves) that help with gas exchange in waterlogged soil.
- Cuticle: a waxy layer on leaves that reduces water loss but is also a very limited site for gas exchange.

How stomata open and close

Guard cells surrounding each stoma change shape to open or close the pore, largely by taking in or losing water — theories explaining this include the photosynthetic theory, the starch-sugar interconversion theory, and the potassium ion theory.

Aerobic vs anaerobic respiration

- Aerobic respiration: uses oxygen to fully break down glucose, releasing a large amount of energy, plus carbon dioxide and water.
- Anaerobic respiration: happens without oxygen, releases much less energy, and produces different by-products (e.g. ethanol and carbon dioxide in plants/yeast).

Economic importance of anaerobic respiration

- Brewing and baking — yeast ferments sugars to produce alcohol and carbon dioxide (bread rising).
- Biogas production from decomposing organic matter.
- Making silage (fermented animal feed) and some fermented foods.



TRY THIS

Mix yeast, sugar, and warm water in a bottle, and fit a balloon over the top. Watch the balloon inflate as the yeast respires anaerobically and produces carbon dioxide.

STRAND 3

ANATOMY AND PHYSIOLOGY OF ANIMALS

From insect mouthparts to the human heart, this strand explores how animals feed, transport materials around their bodies, and exchange gases to stay alive.

3.1 Nutrition in Animals

Different animals have evolved mouthparts and beaks perfectly suited to how they feed.

Insect mouthparts

- Biting and chewing (locust, grasshopper, cockroach): strong mandibles for cutting and grinding plant material.
- Piercing and sucking (mosquito): needle-like mouthparts to pierce skin and suck fluids/blood.
- Siphoning (butterfly/moth): a long, coiled tube (proboscis) for sucking nectar from flowers.
- Cutting (tsetse fly): sharp mouthparts adapted for cutting skin to feed on blood.

Bird beaks

- Short, strong, conical beaks — for cracking seeds and grains.
- Long, thin beaks — for sipping nectar or probing for insects.
- Hooked beaks — for tearing flesh (birds of prey).
- Flat, broad beaks — for filter-feeding (ducks) or catching fish.
- Chisel-like beaks — for chipping wood (woodpeckers).



DID YOU KNOW?

A bird's beak shape is one of the clearest examples of adaptation in nature — you can often guess what a bird eats just by looking at its beak!



TRY THIS

On a nature walk (with a friend, in a safe area), observe two or three different bird species. Note their beak shapes and guess what they might eat, then check if you're right.

3.2 Transport in Animals

Animals need a transport system to move oxygen, nutrients, waste, and hormones around the body — different groups of animals have evolved different systems.

Types of circulatory systems

- Open circulatory system (e.g. insects): blood (haemolymph) flows freely through body spaces, not fully enclosed in vessels.
- Closed circulatory system (e.g. fish, mammals): blood stays within vessels throughout the body.
- Single circulation (fish): blood passes through the heart once per full circuit of the body.
- Double circulation (mammals): blood passes through the heart twice per circuit — once to the lungs, once to the body — allowing more efficient oxygen delivery.

The mammalian heart

The mammalian heart has four chambers (two atria, two ventricles) and pumps blood in two separate loops: to the lungs to collect oxygen (pulmonary circulation), and to the rest of the body to deliver it (systemic circulation).

Lymphatic and immune systems

- The lymphatic system drains excess fluid from tissues back into the bloodstream and helps fight infection.
- The immune system defends the body against pathogens using white blood cells and antibodies.

Blood clotting

When a blood vessel is damaged, platelets and clotting factors work together to form a clot, sealing the wound and preventing excessive blood loss.

Blood groups

- The ABO system classifies blood into groups A, B, AB, and O, based on antigens on red blood cells.
- The Rhesus (Rh) factor further classifies blood as positive or negative.
- Knowing blood group compatibility is vital for safe blood transfusions.



KEY WORDS

Antigen: a marker on a cell that can trigger an immune response.

Platelets: blood cell fragments that help blood clot.

Transfusion: transferring blood from a donor to a recipient.



QUICK TIP

O-negative blood is often called the 'universal donor' because it can be given to patients of nearly any blood group in an emergency.

3.3 Gaseous Exchange and Respiration in Animals

Animals need efficient surfaces to exchange oxygen and carbon dioxide with their environment, and use respiration to release energy from food.

Characteristics of a good respiratory surface

- Large surface area to allow enough gas exchange.
- Thin walls (often one cell thick) to allow gases to diffuse quickly.
- Moist surface to allow gases to dissolve before diffusing.
- Good blood supply (in most animals) to maintain a steep diffusion gradient.

Respiratory structures across animal groups

- Insects: a tracheal system — tiny tubes (tracheae) carry air directly to tissues, opening to the outside through spiracles.
- Fish: gills — thin, feathery structures with a large surface area, richly supplied with blood, extracting oxygen from water.
- Amphibians: use lungs, the moist skin, and the lining of the mouth (buccal cavity) for gas exchange.
- Birds and mammals: lungs with millions of tiny air sacs (alveoli in mammals) to maximise surface area for gas exchange.

Breathing in humans

Inhalation and exhalation are driven by changes in the chest cavity, mainly using the diaphragm and rib muscles, which change air pressure inside the lungs to draw air in or push it out.

Aerobic and anaerobic respiration, and oxygen debt

- During vigorous exercise, muscles may not get enough oxygen and switch to anaerobic respiration, producing lactic acid.
- This creates an 'oxygen debt' — extra oxygen your body needs afterwards to break down the lactic acid, which is why you keep breathing heavily after a sprint.

Respiratory quotient (RQ)

The respiratory quotient compares the volume of carbon dioxide released to the volume of oxygen used during respiration. Different food substrates (carbohydrates, fats, proteins) give different RQ values, which can indicate what type of food is being respired.



TRY THIS

Do 2 minutes of star jumps or running on the spot, then check your breathing rate immediately after and again after resting for a minute. Notice how it changes as your body repays its 'oxygen debt'.



DID YOU KNOW?

A resting adult breathes about 12–16 times a minute — but during intense exercise, this can more than double as your body works to supply enough oxygen to your muscles.